The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

Giulio Ruffini[1,2], Borja Mercadal[1], Ricardo Salvador[1], Alex Just[1], Raul de Palma Aristides[2], Francesca Castaldo[1,2], Nir Grossman[5], Santiago Canals[2,4], and Claudio Mirasso[2,3]

[1]Neuroelectrics, Barcelona, Spain
[2]BCOM, Barcelona, Spain
[3]IFISC (UIB-CSIC), Palma de Mallorca, Spain
[4]Instituto de Neurociencias (UMH-CSIC), Sant Joan d'Alacant, Spain
[5]Department of Brain Sciences and UK Dementia Research Institute, Imperial College London, London, United Kingdom

E-mail: giulio.ruffini@neuroelectrics.com

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ORCID iDs: G. Ruffini 0000-0003-3084-0177; B. Mercadal 0000-0002-0941-6796; R. Salvador 0000-0003-1235-6533; A. Just 0009-0006-8794-7425; R. de Palma Aristides 0000-0002-0255-2293; F. Castaldo 0000-0002-4947-4552; N. Grossman 0000-0003-2002-9049; S. Canals 0000-0003-2175-8139; C. Mirasso 0000-0003-2980-7038.

Supplementary Material

Appendices and supplementary figures accompanying the article. Section, equation and figure numbers referenced here follow the main text; supplementary figures are numbered S1, S2, ...

A Notation and units

Table 1 groups the symbols by their role in the calculation. JR and LaNMM voltages are in mV and rates in s^{-1}. QIF voltages and current-like drives are dimensionless model variables; QIF rates are physical rates, integrated in ms^{-1} and converted to Hz where indicated. A tilde marks the dimensionless membrane-time variables of the uncoupled MPR calculation. All frequencies and time constants in a formula must use the same time unit. A star marks a fixed point or, on a bifurcation parameter, its critical value. Subscripts E and I identify excitatory and inhibitory populations; they are not field or current variables.

Table 1: Notation and units. "Normalized" means dimensionless in the stated model scaling; output units depend on the named observable. Numbered references identify main-text definitions; lettered locations identify appendices (in the supplement for the JNE edition).

Symbol	Meaning	Units	Defined in
Field and access			
$E(t), E_0, \mathbf{E}_j$	Applied field, its amplitude, and vector field of tone $j = 1, 2$	V/m	Eq. (7)
$s(t), \varepsilon_j, \varepsilon$	Local polarization; tone amplitudes; common amplitude ($\varepsilon_{\mathrm{TI}}$ and $\varepsilon_{\mathrm{tACS}}$ label the waveform)	mV	Eqs. (1), (8)
$I_j^{\mathrm{pair}}, \mathcal{P}_j$	Peak electrode-pair current; signed local polarization per unit current	mA; mV/mA	§2.5
$\mathcal{A}(t), \overline{\mathcal{A}}$	Nonnegative polarization envelope and its beat-period mean	mV	Eq. (16); Table 2
f_1, f_2, f_c	Tone frequencies and their center	Hz	Eq. (1)
ω_j, ω_c	Angular frequencies $2\pi f_j, 2\pi f_c$	rad/s	Eqs. (1), (8)
$\Delta f, \Omega$	Beat frequency $f_2 - f_1$; $\Omega = 2\pi\Delta f$	Hz; rad/s	Eq. (1)
m	AM-surrogate modulation depth	dimensionless	Eq. (23)
$\Lambda[E], \lambda_0, \lambda_f$	Field-induced polarization; static and frequency-dependent coupling	mV; mV/(V/m)	Eqs. (3), (8)
$\boldsymbol{\lambda}_0^k, \boldsymbol{\lambda}_f^k$	Coupling vectors for compartment k	mV/(V/m)	Eq. (8)
H, H_k, H_m	Dominant-compartment access transfer, indexed transfer, and point-membrane transfer	dimensionless	Fig. 2; §2.2, §2.8
τ_m, τ	Membrane time; access-filter time in kHz controls	ms	Eqs. (4), (21)
f_{char}	Characteristic frequency $1/(2\pi\tau)$	Hz	Table 3
$\ell(\omega), \ell_0$	Frequency-dependent and static cable space constants	μm	App. B.1, Eq. (4)
$\mathbf{V}, \mathsf{C}_m, \mathsf{G}_m$	Compartment voltages, capacitance and conductance matrices	V; F; S	App. B.1
$\mathbf{I}^0, \mathbf{p}_E$	Endogenous currents; field-to-current vector	A; A/(V/m)	App. B.1
$\mathbf{x}, \phi_j, \mathbf{n}$	Position, field potential, unit cell orientation	m; V; unit vector	§2.5, §2.6
$\mathcal{I}$	Mean-square local polarization $(\varepsilon_1^2 + \varepsilon_2^2)/2$	mV^2	§2.6
Detection			
$\sigma, \sigma', \sigma''$	JR/LaNMM rate curve, slope and curvature	s^{-1}; $s^{-1}mV^{-1}$; $s^{-1}mV^{-2}$	Eq. (2)
e_0, v_0, ρ	Half the rate ceiling, inflection voltage, inverse voltage scale	s^{-1}; mV; mV^{-1}	Eq. (2)
PEIX	Signed sigmoid saturation index	dimensionless	Eq. (22)
$Y_\Omega, Y_{\mathrm{DC}}$	Complex beat amplitude and signed mean shift before recurrence	detector-output units	Eqs. (10), (9)
$\chi^{(2)}$	Dynamic coefficient mapping a polarization product to a rate beat	$s^{-1}mV^{-2}$	§2.2
g, T, Y_{LF}	Memoryless response, envelope transduction, carrier-averaged output	specified output units	Eq. (14)

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Table 1 continued

Symbol	Meaning	Units	Defined in
c_1, c_2, c_4	Linear, quadratic and quartic envelope coefficients	output/(mV)^{1,2,4}	§2.4
$\mathcal{H}_{\mathrm{det}}, \mathcal{H}_{\mathrm{meas}}$	Second/fundamental harmonic magnitude ratios before and after the network	dimensionless	Eq. (15)
$\mathcal{M}[y]$	Peak-to-peak excursion of the quantity y	units of y	Eq. (17)
$F(Q), F', F''$	MPR steady-rate curve and drive derivatives	normalized	Eq. (5)
Q, q	Mean total drive $\bar{\eta} + I$; single-cell drive $\eta + I$	normalized	Eq. (5); App. B.2
$I_{\mathrm{ac}}, \tilde{\omega}$	Single-tone current amplitude; angular frequency $\omega\tau_m$	normalized	§2.3
$\kappa_{\mathrm{DC}}, \kappa_V$	Mean-rate mixing per squared current or squared voltage excursion	normalized	Eqs. (11), (13)
$d, \nu, \tilde{\omega}_z$	MPR relaxation, ringing and zero-crossing angular frequencies	membrane-time units	§2.3
$\mathsf{J}, \mathbf{h}, H_v$	Reduced MPR Jacobian, unit-current response and voltage transfer	normalized	§2.3; App. B.2
$\mathbf{b}_2, \mathsf{I}_2$	Quadratic DC source per I_{ac}^2; identity matrix	normalized	App. B.2
$\mathcal{R}_{\mathrm{det}}, \mathcal{R}_{\mathrm{lin}}$	TI/direct-drive source ratio; linear TI/tACS output ratio	dimensionless	App. D; §4.8
δ, ζ	Threshold offset; sigmoid rate divided by $2e_0$	mV; dimensionless	§2.4; App. D
$u, \xi, \beta, k_{\mathrm{ell}}^2$	Reduced amplitude $\rho\varepsilon$, offset $\rho(v^* - v_0)$, threshold ratio, elliptic parameter	dimensionless	App. D
$L, \psi, \mathsf{K}, \mathsf{E}$	Unit logistic, normalized waveform, complete elliptic integrals	dimensionless	App. D
Network			
r_X, v_X	Population firing rate and mean potential, indexed by X	s^{-1}; mV in JR/LaNMM, normalized in QIF	§2.1
$\tilde{t}, \tilde{r}, \tilde{f}$	Membrane-time coordinate, population rate and single-cell rate	normalized	Eq. (4); App. B.2
$r_{\mathrm{syn}}, r_E^{\mathrm{syn}}, r_I^{\mathrm{syn}}$	Synaptically filtered firing rates	same rate units as input	Eqs. (6), (25)
$u_{\mathrm{PSP}}, u_{X\leftarrow Y}, \hat{L}$	Postsynaptic potential and inverse synaptic operator	mV; maps mV to s^{-1}	Eq. (3)
A, B, a, b	JR excitatory/inhibitory PSP gains and inverse synaptic times	mV; s^{-1}	§2.1; App. B
h_e, h_i	JR excitatory and inhibitory synaptic impulse responses	mV	Fig. 2
$C, C_{1\ldots4}, C_{X\leftarrow Y}$	JR connectivity scale and pathway weights	dimensionless	§2.1; App. B
y_0, y_1, y_2, v	JR PSPs; voltage readout $v = y_1 - y_2$	mV	App. B
p, p_H, C^*	JR external rate and its upper Hopf value; critical connectivity	s^{-1}; dimensionless for C^*	§3.4, §4.6
$\eta_j, \bar{\eta}, \Delta$	QIF excitabilities, Lorentzian center and half-width	normalized	Eq. (4)
$I, I_E, I_j, A_f, \lambda_I$	QIF input, E-population input, tone amplitudes, forcing amplitude and field coupling	normalized; λ_I in (V/m)$^{-1}$	Eqs. (4), (25)
$\mathcal{U}_j$	Total synaptic and field drive to QIF neuron j	normalized	Fig. 2
J, J^*, w_{XY}	QIF coupling, critical coupling and relative pathway weights	dimensionless	Eq. (25)
$\tau_E, \tau_I, \tau_a, \tau_g, \tau_s$	E/I membrane and excitatory/inhibitory/generic synaptic times	ms	Eqs. (6), (25)
$\hat{K}_a, \hat{K}_g, H_s$	QIF synaptic rate filters; JR rate-to-voltage transfer	dimensionless; mV s for H_s	§3.5, §2.7

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Table 1 continued

Symbol	Meaning	Units	Defined in
$\mu_{\pm}, \gamma, \omega_0, f_0$	Modal eigenvalues, damping, angular and ordinary modal frequency	s^{-1} (ms^{-1} where stated); Hz for f_0	Eq. (19)
$\tau_{\mathrm{rel}}, \tau_{\mathrm{rev}}, \mathcal{Q}$	Population relaxation time, $1/\gamma$, and mode quality factor	time; dimensionless for $\mathcal{Q}$	§2.1, §2.7, §5
D, G, R_Ω	Beat-source coefficient, recurrent transfer, complex observed beat response	input/output-dependent	Eq. (20)
$V_j, V_{\mathrm{peak}}, N$	Single-QIF voltage, spike cutoff and neurons per population	normalized; count	§3.6
$\delta\eta_E, \delta\eta_E^*$	Rectified E-drive shift and its Hopf threshold	normalized	§3.6, §4.10
P_1, P_2, r_P, φ_X	Deep/superficial pyramidal populations; JR and LaNMM pyramidal rates	labels; s^{-1} for rates	§2.1; §E.1
$z, \alpha, \mathcal{F}, \varepsilon_{\mathrm{SL}}$	Stuart–Landau mode, growth rate, forcing and forcing amplitude	normalized mode; s^{-1} for the others	App. C
Measurement			
$L_{\cos}, L_{\sin}, A_\Omega$	Signed lock-in quadratures and nonnegative amplitude	measured voltage or rate units	Eq. (24)
A_{open}	Magnitude of the open-loop rate beat	s^{-1}	§4.6
$w_k, t_k, v_k, \bar{v}$	Hann weights, sample times, voltage samples and weighted mean	1; s; mV; mV	Eq. (24)
$A_{\mathrm{test}}, \theta_0$	Amplitude and phase of the lock-in test tone	readout units; rad	§3.2
$\Delta t, N_{\mathrm{grid}}$	Integration step and number of parameter combinations	s or ms; count	§3.4
$A_\Omega/\langle r_E \rangle$	Beat-locked population modulation depth	dimensionless	§3.6
PPC	Pairwise phase consistency of single-neuron spike phases	dimensionless	§3.6
$f_{\mathrm{out}}, S(f_c)$	Dominant output frequency; LaNMM mean off-alpha lock-in	Hz; mV	App. F; §E.1

B Simulated models: equations and parameters

The models are written in the NMM1 transfer-function formalism of Eq. (3) [1]: synaptic operators $\hat{L}$ transduce firing rates into PSPs, the PSPs sum with any direct field term $\Lambda[E]$ in the sigmoid argument, and the sigmoid returns the rate. The Jansen–Rit column is given here; the laminar model is specified in [2, 3] with its field drive in Eq. (9), and the NMM2 PING motif and its QIF network in Eq. (25) and §3.6.

Jansen–Rit column

Writing y_0 for the excitatory PSP driving the interneurons, y_1 and y_2 for the excitatory and inhibitory PSPs onto the pyramidal population, external input p, and the applied-field perturbation $s(t)$ (§2.2) entering the pyramidal sigmoid,

$$\ddot{y}_0 = Aa\,\sigma\big(y_1 - y_2 + s(t)\big) - 2a\dot{y}_0 - a^2 y_0, \tag{1}$$

$$\ddot{y}_1 = Aa\,\big[\,p + C_2\,\sigma(C_1 y_0)\,\big] - 2a\dot{y}_1 - a^2 y_1, \tag{2}$$

$$\ddot{y}_2 = Bb\,C_4\,\sigma(C_3 y_0) - 2b\dot{y}_2 - b^2 y_2, \tag{3}$$

integrated as six first-order ODEs; the LFP proxy is $v = y_1 - y_2$ [4]. Table 2 gives the reference column parameters. The synaptic rates correspond to excitatory and inhibitory decay times of 10 and 20 ms, while A and B set the postsynaptic-potential gains. The sigmoid parameters fix its voltage scale and rate ceiling; the sweeps vary the external rate p or connectivity C to change the operating state, as specified in Methods.

Table 2: Jansen–Rit parameters [4].

Symbol	Meaning	Value
A, B	excit./inhib. PSP gain	3.25, 22 mV
a, b	synaptic rates	100, 50 s^{-1}
v_0, e_0, ρ	sigmoid inflection, max rate/2, slope	6 mV, 2.5 s^{-1}, 0.56 mV^{-1}
C	global connectivity	135
C_1, C_2, C_3, C_4	couplings	C, $0.8C$, $0.25C$, $0.25C$
p	external input (bifurcation knob)	0–400 Hz

B.1 Compartment access and the passive-cable estimate

The passive compartment equation $\mathsf{C}_m\dot{\mathbf{V}} + \mathsf{G}_m\mathbf{V} = \mathbf{I}^0 - \mathbf{p}_E E$ [5] gives the voltage perturbation $\delta\mathbf{V}(\omega) = -(\mathsf{G}_m + i\omega\mathsf{C}_m)^{-1}\mathbf{p}_E E(\omega)$, where $i^2 = -1$. In SI units, $\mathbf{V}$ contains the compartment membrane potentials (V), C_m and G_m are the capacitance (F) and conductance (S) matrices, $\mathbf{I}^0$ is the endogenous current (A), and $\mathbf{p}_E$ is the field-to-current coupling (A per V/m).

Geometry can also alter the relevant response mode. For a passive cable, the magnitude of the frequency-dependent space constant is

$$\ell(\omega) = \ell_0(1 + \omega^2\tau_m^2)^{-1/4}, \tag{4}$$

where ℓ_0 is its static value. At 1 kHz and $\tau_m = 16$ ms, this length is about one tenth of its static value, so compartments that are strongly coupled at low frequency can respond differently to the carrier. A somatic charging time alone therefore does not determine every extracellular-field response mode. For a sufficiently long passive cable with a sealed end in a uniform axial field, the end polarization is proportional to this space constant. Each carrier then acquires the factor $(1 + \omega^2\tau_m^2)^{-1/4}$, so the quadratic beat is attenuated by $(1 + \omega_c^2\tau_m^2)^{-1/2}$, about 10^{-2} at 1 kHz and 10^{-3} at 10 kHz for 16 ms. These differ from the point-membrane estimates; soma loading and finite cable length further change the transfer. These distinct modes motivate the compartment-specific access hypothesis; their implications for TI are considered in §5.

B.2 Single-cell rate and dynamic MPR mixing

The MPR steady-rate curve F of §2.1, Eq. (5), averages the single-neuron rate $\tilde{f}(q) = \sqrt{q}/\pi$ for $q > 0$, and $\tilde{f}(q) = 0$ otherwise, over the excitability distribution. Here η is the neuron's excitability and $q = \eta + I$ its constant total drive.

The coefficient in Eq. (11) follows by expanding about the fixed point $(\tilde{r}^*, v^*)$, with $v^* = -d/2$, in membrane-time units. The Jacobian and unit-current phasor are

$$\mathsf{J} = \begin{pmatrix} 2v^* & 2\tilde{r}^* \\ -2\pi^2\tilde{r}^* & 2v^* \end{pmatrix}, \qquad \mathbf{h} = (i\tilde{\omega}\mathsf{I}_2 - \mathsf{J})^{-1}\begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

where I_2 is the identity matrix and the first-order response is $I_{\mathrm{ac}}\,\mathrm{Re}(\mathbf{h}e^{i\tilde{\omega}\tilde{t}})$. Writing h_r, h_v for its rate and voltage components, with an overbar denoting complex conjugation and Re the real part, the carrier-averaged quadratic source per I_{ac}^2 is

$$\mathbf{b}_2 = \begin{pmatrix} \mathrm{Re}(h_r\overline{h_v}) \\ (|h_v|^2 - \pi^2|h_r|^2)/2 \end{pmatrix}.$$

The mean second-order correction is $-I_{\mathrm{ac}}^2\mathsf{J}^{-1}\mathbf{b}_2$; its rate component gives Eq. (11).

C The cubic Hopf normal form lacks a quadratic mixer

The Stuart–Landau equation [6]

$$\dot{z} = (\alpha + i\omega_0)z - |z|^2 z + \mathcal{F}(t), \qquad \gamma = -\alpha > 0, \tag{5}$$

uses seconds for time and a normalized complex mode z, with the cubic coefficient set to 1 s^{-1}. Its linear growth rate is $\alpha = -\gamma$; γ has units s^{-1}. The mode has damping γ and natural angular frequency ω_0, driven by the applied input $\mathcal{F}$ (normalized mode units per second). For a weak input $\mathcal{F} = \varepsilon_{\mathrm{SL}}\psi_{\mathrm{SL}}(t)$ with dimensionless waveform ψ_{SL} on the stable side, the perturbative response has the form $z = \varepsilon_{\mathrm{SL}} z_1 + \varepsilon_{\mathrm{SL}}^3 z_3 + \cdots$. There is no second-order source: the first-order response contains only the applied frequencies, and the cubic first contributes at order $\varepsilon_{\mathrm{SL}}^3$. Thus the normal form has a resonant linear susceptibility but lacks the quadratic difference-frequency generation of Eq. (10). This statement concerns the weak-input expansion; it is not an exclusion of every higher-order intermodulation or strong-drive response.

In the numerical control ($\alpha = -0.5$ s^{-1}, $f_0 = 10$ Hz, $f_c = 100$ Hz, $\varepsilon_{\mathrm{SL}} = 0.5$ s^{-1}), a square-law detector gives a beat-frequency lock-in amplitude of 2.5×10^{-1}, while $\mathrm{Re}\, z$ gives approximately 2×10^{-8}, at the numerical floor. Driving the same mode directly in band recovers the forced $1/\gamma$ growth. The control separates resonant amplification from the nonlinear source required to drive it at the beat.

D Efficiency of a memoryless transfer against a direct in-band drive

For the logistic family of memoryless transfers, compare balanced TI tones of polarization amplitude ε each with a direct in-band drive of amplitude 2ε, matching peak polarization. Their leading drive amplitudes are $|\sigma''|\varepsilon^2/2$ and $2\sigma'\varepsilon$, so $\mathcal{R}_{\mathrm{det}} \simeq |\sigma''|\varepsilon/(4\sigma')$. The weak-input formula concerns the logistic family. The step-mixture argument below extends the exact carrier-averaged bound to bounded nondecreasing memoryless transfers, with the same operating point, matched peak polarization, and open-loop readout. It does not cover the dynamic detector of §2.3.

Writing $\zeta = \sigma/(2e_0) \in (0,1)$ gives $\sigma' = 2e_0\rho\zeta(1-\zeta)$ and $\sigma'' = 2e_0\rho^2\zeta(1-\zeta)(1-2\zeta)$. Since $|\mathrm{PEIX}| = 1 - 4\zeta(1-\zeta) = (1-2\zeta)^2$, the leading ratio satisfies

$$\mathcal{R}_{\mathrm{det}} \simeq \frac{\rho\,\varepsilon}{4}\sqrt{|\mathrm{PEIX}|} \;\le\; \frac{\rho\,\varepsilon}{4}. \tag{6}$$

It vanishes at the inflection, where the quadratic coefficient is zero, and approaches its bound in the saturated tails. This maximizes a ratio rather than an absolute response: both drives vanish in those tails. For the JR slope $\rho = 0.56\ \mathrm{mV}^{-1}$, the perturbative bound is 1.4% at $\varepsilon = 0.1$ mV and 0.28% at $\varepsilon = 0.02$ mV.

The slope is itself a parameter, so the question is what the whole sigmoid family can do. The logistic transfer depends on voltage only through $\rho(v - v_0)$, so writing

$$u = \rho\,\varepsilon, \qquad \xi = \rho\,(v^* - v_0), \tag{7}$$

we have $\sigma(v^* + \varepsilon\psi(t)) = 2e_0\,L(\xi + u\,\psi(t))$ for any normalized waveform ψ, with $L(\xi) = (1+e^{-\xi})^{-1}$. Every Fourier amplitude, and hence their ratio, is a function of (u,ξ) alone; the factor $2e_0$ cancels. Maximizing over v^* and ρ at fixed ε is therefore the same as maximizing over (u,ξ), and the family supremum is independent of field strength. Evaluating the exact cycle response rather than its quadratic truncation and optimizing ξ at each u, the ratio rises monotonically over the range explored and converges to the hard-threshold limit

$$\sup_{\rho,\,v^*} \mathcal{R}_{\mathrm{det}} \approx 0.3479, \tag{8}$$

approached as $\rho \to \infty$ at a threshold offset δ (the distance from the operating point to the inflection) with $\delta/2\varepsilon = 0.523$. There is no finite optimal transfer width: narrower is always better, and reaching 95% of the ceiling requires $u \gtrsim 6$, that is $1/\rho \lesssim \varepsilon/6$.

Such a detector is not weak. At the optimum, and at the illustrative fast-access polarization $\varepsilon \simeq 0.1$ mV of §5, its open-loop beat drive exceeds the physiological-slope sigmoid's by some three orders of magnitude for the same e_0 and waveform, because the field excursion is about twice the threshold offset and the transfer switches across its full range instead of responding linearly. The disparity grows as the field weakens, the smooth sigmoid's response falling as ε^2 while a threshold repositioned with the field keeps its excursion.

The obstacle is the width. At $\varepsilon = 0.1$ mV the ceiling is approached only for $1/\rho \lesssim 17\ \mu$V; at $\varepsilon = 0.02$ mV the corresponding width is about 3 μV. Both follow from $1/\rho \lesssim \varepsilon/6$, with the threshold offset held to the same order as the applied polarization. A population sigmoid averages the spread of thresholds and noise; the JR voltage scale is $1/\rho \approx 1.8$ mV [4]. A microvolt width lies far below that and reads as an idealized microscopic threshold or rectifier rather than as a neural-mass sigmoid; real thresholds are themselves noisy and state-dependent, so it is not a measured single-cell property either. The regime in which the family approaches its ceiling is thus one in which the "population sigmoid" has ceased to describe a population.

The limit is a step detector, not an envelope follower. For a unit step at offset δ the carrier-averaged transduction is $\pi^{-1}\arccos(\delta/\mathcal{A})$ for $\mathcal{A} > \delta$ and zero below; with $\beta = \delta/2\varepsilon$ its tACS fundamental is $(2/\pi)\sqrt{1-\beta^2}$ and its TI coefficient reduces by parts to $\mathcal{R}_{\mathrm{det}} = 2\beta\,[\mathsf{K}(k_{\mathrm{ell}}^2) - \mathsf{E}(k_{\mathrm{ell}}^2)]/(\pi\sqrt{1-\beta^2})$

with elliptic parameter $k_{\mathrm{ell}}^2 = 1 - \beta^2$ and K, E the complete elliptic integrals, giving 0.3479 at $\beta = 0.523$ by direct quadrature and by the elliptic form alike (`threshold_bound_check.py`). At that optimum the step's second-to-first harmonic magnitude ratio is 0.151, not the 1/5 of a linear envelope follower. A bounded nondecreasing transfer is, up to a constant, a positive mixture of shifted steps. Both signed Fourier coefficients are linear in the transfer, and each step's direct-drive coefficient is nonnegative. The triangle inequality therefore bounds the mixture's amplitude ratio by the largest step ratio, whenever the denominator is nonzero. This extends the result beyond the logistic family; the maximum itself is a numerical evaluation, not a certified bound.

E Extension: the laminar column at low carriers

The two-band laminar column (LaNMM, §2.1) adds a resonant intermediary between the field and the slow generator. These sweeps use 25–80 Hz carriers and polarizations up to 3 mV on P_2, exceeding the sigmoid scale $1/\rho \approx 1.8$ mV. They describe nonlinear frequency selection below TI's kHz range. The weak-field square law is tested separately in JR at 0.05–0.2 mV, one to two orders of magnitude below that scale (Fig. 7b).

E.1 Model and measurements

We complement the single-column analysis with simulations of the laminar neural mass model (LaNMM) of §2.1. We integrate the full 28-variable column [2, 3], with the field entering as in the single JR column, as a membrane polarization added inside the sigmoid argument of the target population. Its output rate φ_X (s^{-1}) is

$$\varphi_X(t) = \sigma\big(v_X(t) + s(t)\big), \qquad X \in \{P_1, P_2\}, \tag{9}$$

with $s(t)$ the AM or two-tone field of §2.2 (amplitude ε in mV, beat $\Omega = 2\pi\Delta f$, carrier $\omega_c = 2\pi f_c$) and all three external drives held flat at their baselines (Fig. S2). A perturbation of an external input *rate* would require clipping at zero once its excursion exceeds the baseline, and a hard clip is a half-wave rectifier—a diode detector placed upstream of the sigmoid, which would inject a line at Δf into the drive before the population nonlinearity received it. A signed membrane polarization carries no such constraint and none is applied. Because the LaNMM and JR sigmoids share the voltage scale $1/\rho \approx 1.8$ mV, ε is directly comparable between the two models.

Integration is a vectorized fixed-step RK4 ($\Delta t = 10^{-4}$ s) advancing the whole parameter grid simultaneously, as in the JR sweeps. The maps report the alpha-band (8–12 Hz) power of the P_1 and P_2 membrane potentials as the mean squared magnitude of the Hann-windowed fast Fourier transform (FFT) over the band, in arbitrary units, over a post-transient window. This measure includes autonomous and driven activity and does not isolate a phase-coherent response to stimulation.

Two sweeps probe the column: an amplitude–beat sweep over $(\varepsilon, \Delta f)$ at fixed carrier, and a carrier–beat sweep over $(f_c, \Delta f)$ at fixed ε, built on the LaNMM module of [2]. The maps (Figs. S1, S7) show the *signed* stimulation-induced change, alpha power minus the unstimulated ($\varepsilon = 0$) baseline, on a zero-centered scale; over most of the P_1-driven grids the change is negative. For the alpha resonance we instead drive the deep P_1 loop directly and sweep its input against Δf, reading the lock-in (Eq. (24)) of v_{P_1} at Δf.

The additional carrier comparison below uses the lock-in of v_{P_1}, averaged at each carrier over 17 beats: 7–9 and 12.25–14 Hz in 0.25 Hz steps. These lie more than 1.5 Hz from the field-free alpha peak near 10.6 Hz. We denote this arithmetic mean by $S(f_c)$. The carrier grid is 25–80 Hz in 1 Hz steps, with equal tone amplitudes $\varepsilon = 1.5$ mV for P_2 drive and 0.5 mV for P_1 drive. After 2 s of settling, we compare 3 and 10 s measurement windows. Field-free controls give mean background amplitudes of 0.0048 and 0.0012 mV over the same beats. This background includes a persistent component near 8 Hz, so it is not solely finite-window leakage. The statistic compares carrier profiles within each drive condition; the unequal amplitudes do not support a direct sensitivity comparison between targets.

E.2 Frequency selection and carrier dependence

Driving either population with the two-tone field of Eq. (9) and reading alpha-band power in the deep (P_1, alpha) and superficial (P_2, gamma) pyramidal potentials recovers the beat resonance (§E.1, Fig. S2). Direct P_1 drive reproduces the JR alpha-band resonance on both stable-focus and limit-cycle sides; that sweep and its code are in the repository. The additional result is carrier selectivity through the coupled gamma circuit.

Table 3: LaNMM carrier profiles measured away from alpha. $S(f_c)$ is the mean lock-in amplitude of v_{P_1} over the 17 beats specified in §E.1. Percentages are normalized to each row's own carrier maximum; the range and median use carriers from 30 to 80 Hz.

Drive	Window	Peak S (mV)	Peak f_c (Hz)	Range (%)	Median (%)
P_2, 1.5 mV	3 s	0.450	39	23–100	42
P_2, 1.5 mV	10 s	0.450	39	23–100	42
P_1, 0.5 mV	3 s	0.092	25	42–68	53
P_1, 0.5 mV	10 s	0.091	25	42–67	52

Sweeping the field amplitude ε and beat frequency Δf at a fixed carrier produces a region of enhanced alpha-band power: the response in P_1 is largest when Δf matches the deep alpha ($\sim$10 Hz) and grows with ε (Figs. S1a,b and S7a,b). The dominant-frequency contours identify where the output is close to the beat, without establishing phase locking of the autonomous rhythm. Away from the enhancement region the drive *reduces* alpha-band power: with the field on P_1, 95% of the amplitude–beat grid and 97% of the carrier–beat grid lie below the field-free baseline (Fig. S7), the drive pulling the autonomous alpha out of the 8–12 Hz band or suppressing it; with the field on P_2 the fractions are 44% and 31% (Fig. S1).

The carrier dependence illustrates the two coupling limits of §2.2–§2.8 within this model's 25–80 Hz carrier range. When the field is delivered to the superficial loop (P_2), alpha-band power enhancement in P_1 is largest in the gamma carrier band with Δf near the P_1 alpha (Fig. S1c,d). At $\Delta f = 10$ Hz, the carrier maximum is 44 Hz. This selectivity is consistent with a double resonance: the fast loop acts as a tuned radio-frequency (RF) front end whose dynamics affect the carrier dependence of the deep response. With direct P_1 drive, a ridge near the column's own alpha persists across the carrier range (Fig. S7c). That ridge contains autonomous activity and is not a measurement of the demodulated component alone.

The lock-in also contains the autonomous rhythm when the beat is close to alpha; its maximum over beats cannot isolate the driven response. We therefore compare the off-alpha statistic $S(f_c)$ defined in §E.1 (Table 3). With P_2 drive it peaks near 39 Hz and falls to about 16–17% of that peak at a 25 Hz carrier. With P_1 drive it peaks at 25 Hz and spans about 42–68% of that peak over 30–80 Hz. These are properties of the specified amplitudes and beat average, not universal carrier-rejection factors. Single-beat profiles have deeper minima, including a dip near a 45 Hz carrier under direct P_1 drive: the gamma circuit can affect the response in either stimulation condition.

The carrier peaks and aggregate ranges persist with the longer window, although individual normalized values change by up to 3.1 percentage points for P_2 drive and 9.6 for P_1 drive. The smallest S exceeds its mean field-free background by factors of about 16 and 8, respectively, with the 3 s window, and 62 and 32 with 10 s. Some individual beat–carrier points remain near or below background; the comparison concerns the mean over beats, not each grid point. TI's kHz carriers lie outside this model's range. Within it, the carrier can excite a network mode and affect the detected response, qualifying the inert-carrier limit of §2.7 and linking it to frequency-tagging and intermodulation paradigms [7].

Within the 25–80 Hz carrier range and at the finite polarizations used here, the LaNMM shows alpha-band power changes concentrated near the intrinsic alpha, amplitude-dependent widening of the frequency-dominance region, and enhanced responses when superficial drive passes through the gamma circuit. The power maps include autonomous and driven activity; they do not isolate TI efficacy. The gamma-carrier preference, around 40 Hz rather than kHz, remains a separate prediction to test. Esmaeilpour et al. [8] swept 0.1–2 kHz carriers at a fixed 5 Hz beat against carbachol-induced gamma, finding decreasing sensitivity with carrier frequency. Those carriers did not sample the gamma band. Violante et al. [9] also used a fixed 5 Hz beat. Neither study tests

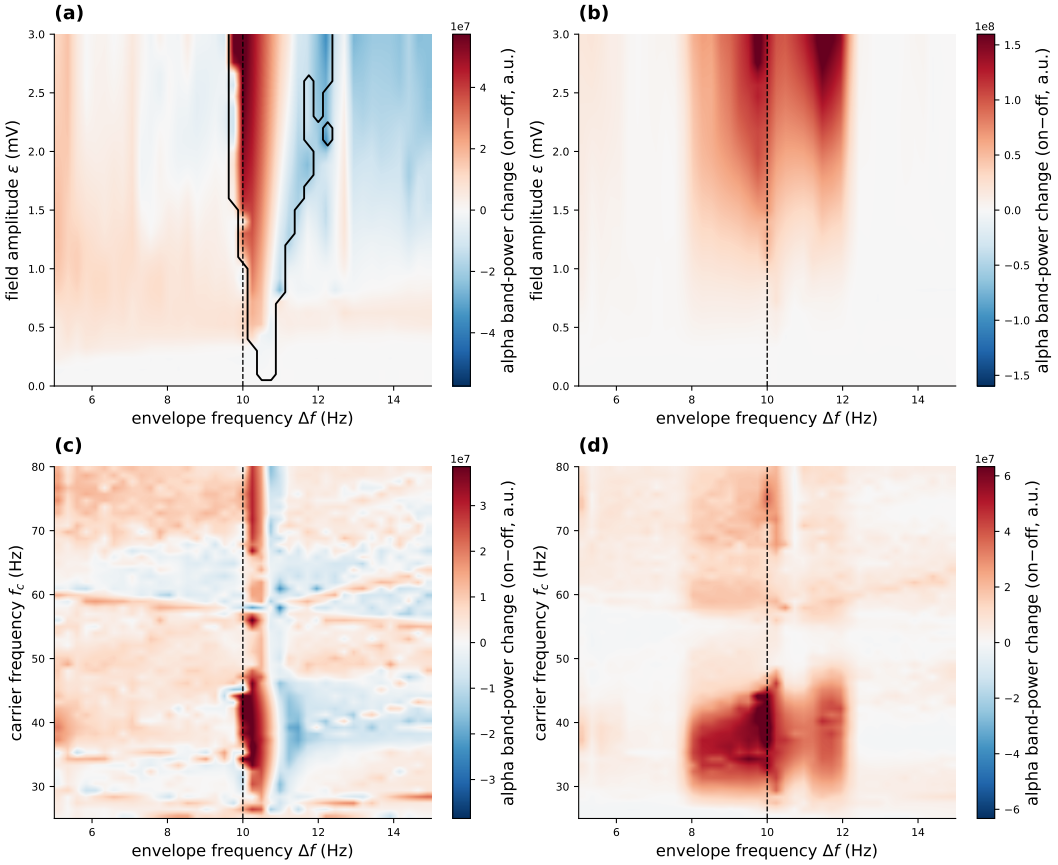


Figure S1: LaNMM frequency selection: two-tone TI field on the superficial P_2 (gamma) loop. The field enters the P_2 sigmoid argument as a membrane polarization ε (mV) with all external drives flat. Color shows the signed change of alpha-band (8–12 Hz) power from the unstimulated $\varepsilon = 0$ baseline, on a zero-centered scale (arbitrary units). *(a,b)* P_1 and P_2 enhancement vs. ε and Δf (carrier 40 Hz), peaking at the P_1 alpha (10.0 Hz, dashed). The black contour in *(a)* is the frequency-dominance set $|f_{\text{out}} - \Delta f| < 0.3$ Hz (Appendix F), drawn because band power alone cannot separate a driven from an autonomous rhythm; it is a dominance screen, not a phase-capture test. The $\varepsilon = 0$ row is excluded, since with no drive the autonomous 10.67 Hz alpha already lies within the tolerance of two grid points (a resolution floor of 0.5 Hz); the set widens to 2.75 Hz at 3 mV. *(c,d)* Enhancement vs. carrier and Δf at $\varepsilon = 1.5$ mV: P_1 alpha-band enhancement is largest in the gamma carrier band with Δf near the P_1 alpha. At $\Delta f = 10$ Hz, the carrier maximum is 44 Hz. The field-free P_2 gamma peak is near 39.7 Hz; the finite-drive maximum need not coincide with it.

this low-carrier resonance prediction.

F Vocabulary and metrics

Forced response. Below the Hopf the column does not oscillate on its own; we quantify its response by the lock-in amplitude at the beat Δf (Eq. (24)). There is no autonomous rhythm to entrain in the stable-focus J-curve, timing, and tACS sweeps.

Frequency dominance and entrainment. Above the Hopf the column free-runs at f_0. A 1:1 phase-capture test would follow the unwrapped phase difference from the drive and require absence of secular drift. Our maps instead report $|f_{\text{out}} - \Delta f| < 0.3$ Hz, where f_{out} is the largest non-DC FFT peak. This is a frequency-dominance screen: a forced component can outgrow a persisting autonomous one without capturing it. When the components coexist, instantaneous-phase estimates also require their separation [10]. The maps therefore identify a dominance region, not a measured Arnold tongue.

Beat-locked rate and gating. Gating means that the amplitude of a persisting rhythm is modulated by the beat without frequency capture [11, 12]. The QIF statistic used here is the lock-in of the population rate r_E at Δf, complemented by the rate folded on beat phase over whole cycles. It measures beat-locked firing. A raw rate fold does not extract the gamma amplitude, and sidebands can arise from amplitude or phase modulation. The above-Hopf QIF panels therefore establish coexistence and mixing of the beat and gamma rhythms; a gamma-amplitude estimate would be needed to identify gating specifically.

The protocol for Fig. S4 places the column on the limit-cycle side (input p on the stable alpha-cycle branch below its upper Hopf), drives it with $s(t) = \varepsilon \cos \Omega t$, and records the dominant frequency. Grid widths and the clipping check are specified in the code. As the Hopf is approached, the autonomous cycle shrinks and the frequency-dominance region broadens in these simulations. Its width and the stable-focus susceptibility describe different responses; their increase near onset does not make them the same quantity.

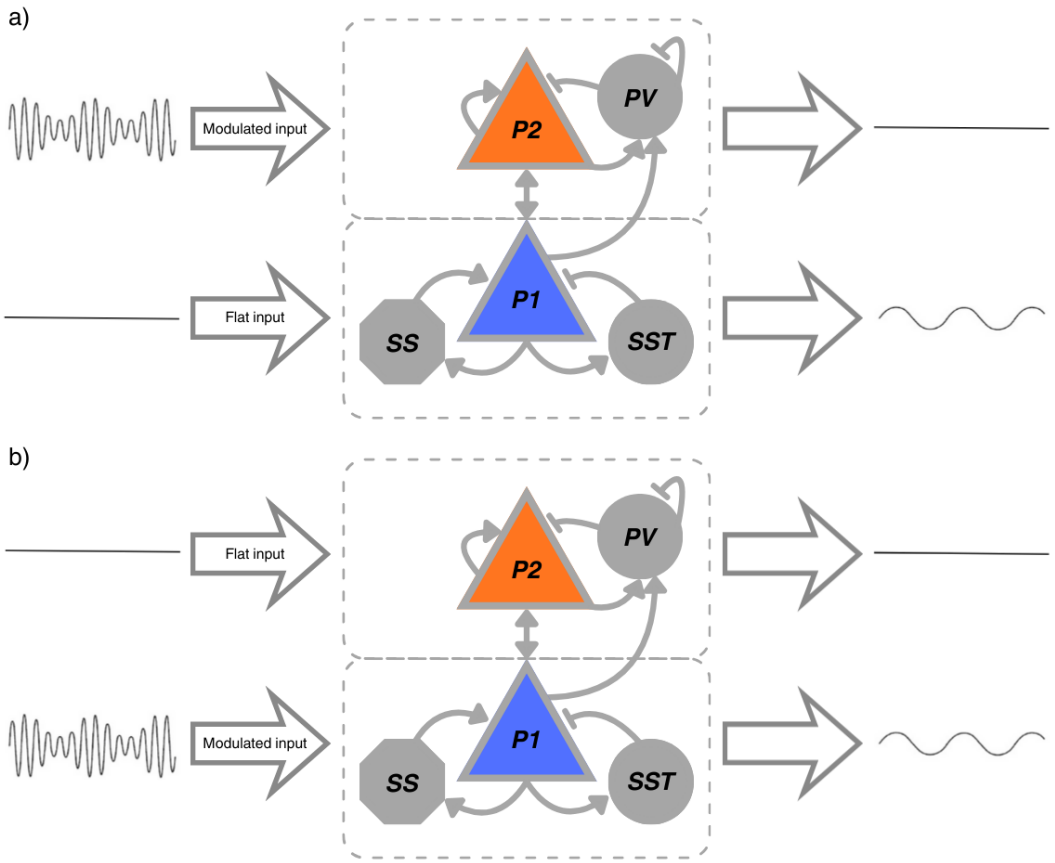


Figure S2: Stimulation setup in a LaNMM column. The column couples a superficial PING subcircuit (pyramidal P_2 + PV; gamma) and a deep JR subcircuit (pyramidal P_1 + SS + SST; alpha). *(a)* the TI field drives P_2 while P_1 receives a flat drive; *(b)* the TI field drives P_1 while P_2 is flat. Through the sigmoid nonlinearities and P_1–P_2 coupling, the generated beat-frequency component is transferred to the deep loop: P_2 carries the modulated fast carrier, while P_1 exposes the recovered low-frequency component (right traces).

G Supplementary figures

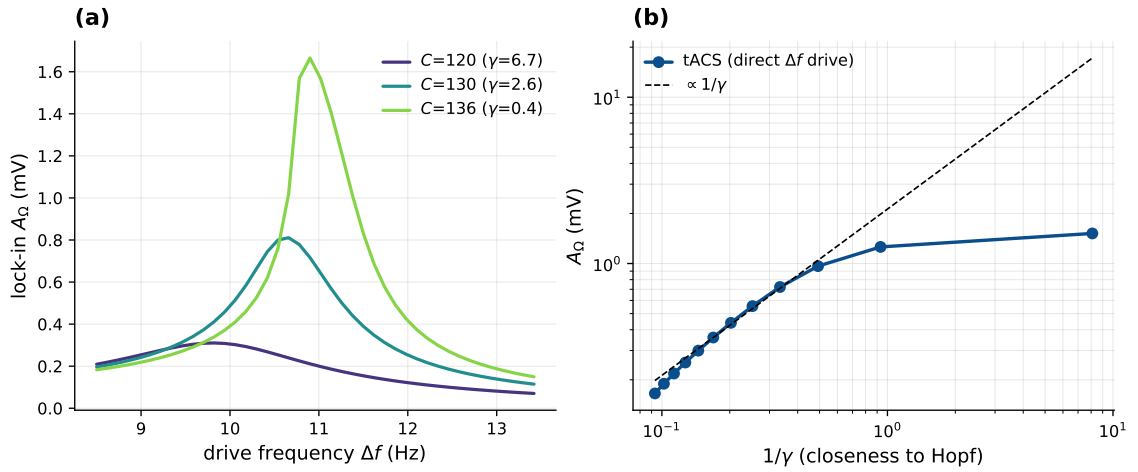


Figure S3: tACS shares the network amplifier (Jansen–Rit). A direct low-frequency (tACS) drive at Δf enters the column without a carrier, so it needs no demodulation, but feeds the same near-Hopf resonator. With the field-free operating point pinned, sweeping the coupling C toward the alpha Hopf: *(a)* the lock-in resonance at Δf sharpens and grows as $C \to C^*$; *(b)* the response grows as $1/\gamma$ (dashed) and saturates near criticality—the same network amplification as the demodulated TI drive (Fig. 9). The leading tACS response is linear in the field; the TI beat is quadratic. Their response ratio is independent of damping in linear response and changes at the finite amplitudes nearest the Hopf (§4.6). Here $\gamma > 0$ describes the field-free state.

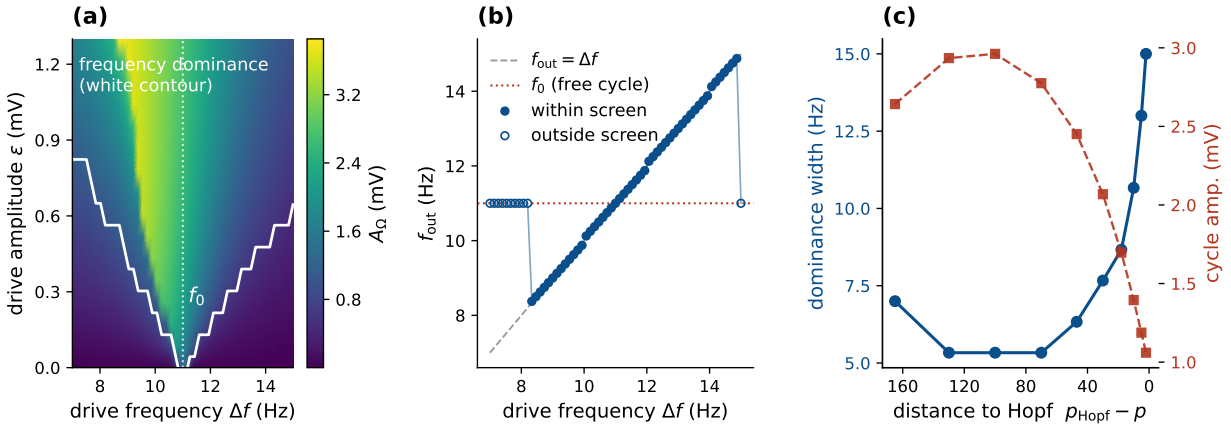


Figure S4: Frequency dominance in the driven autonomous Jansen–Rit column. The column is set in its autonomous alpha limit cycle ($p = 250$, $f_0 \approx 11$ Hz) and driven by a direct in-band sinusoid. *(a)* The set $|f_{out} - \Delta f| < 0.3$ Hz forms a wedge near f_0 that widens with drive. It screens for phase capture but does not establish it. *(b)* Inside this region the dominant frequency follows Δf; outside, it remains near f_0. *(c)* Toward the Hopf at fixed drive the dominance width grows as the autonomous cycle amplitude vanishes (blue, left axis; red dashed, right axis). Above the Hopf this width describes frequency selection in a driven autonomous oscillator; it is distinct from the linear susceptibility of the stable-focus side (Fig. 9). Both diagnostics increase toward criticality in these sweeps.

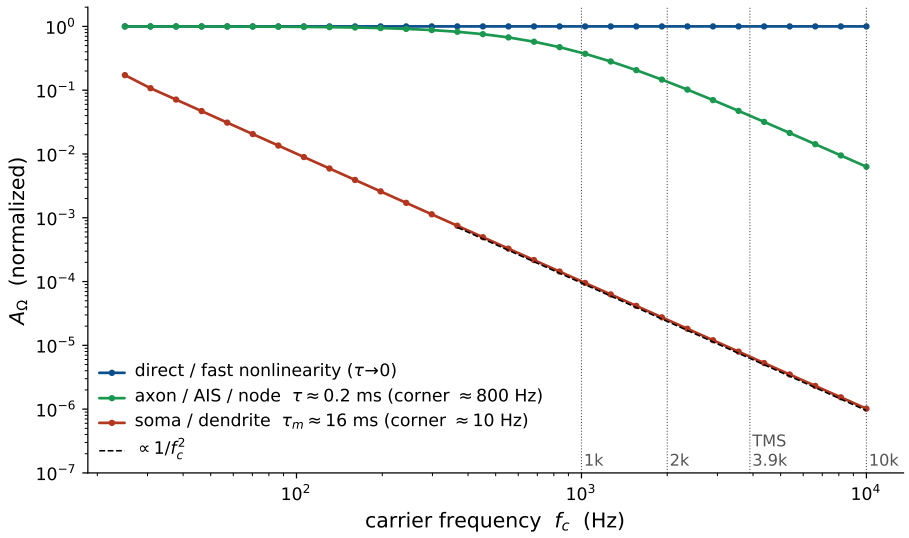


Figure S5: Carrier attenuation depends on the assumed access time (Jansen–Rit, open loop). Open-loop demodulated response (normalized to the $\tau \to 0$ value) vs. carrier frequency up to 10 kHz. With direct coupling ($\tau \to 0$) the response is flat—carrier-independent. A slow somatic membrane ($\tau_m \approx 16$ ms) rolls off as $1/f_c^2$ (dashed guide); the assumed fast response ($\tau = 0.2$ ms) retains about 0.14 of the direct response at 2 kHz and 0.04 at 3.9 kHz. These are sensitivity cases, not measurements of anatomical compartments.

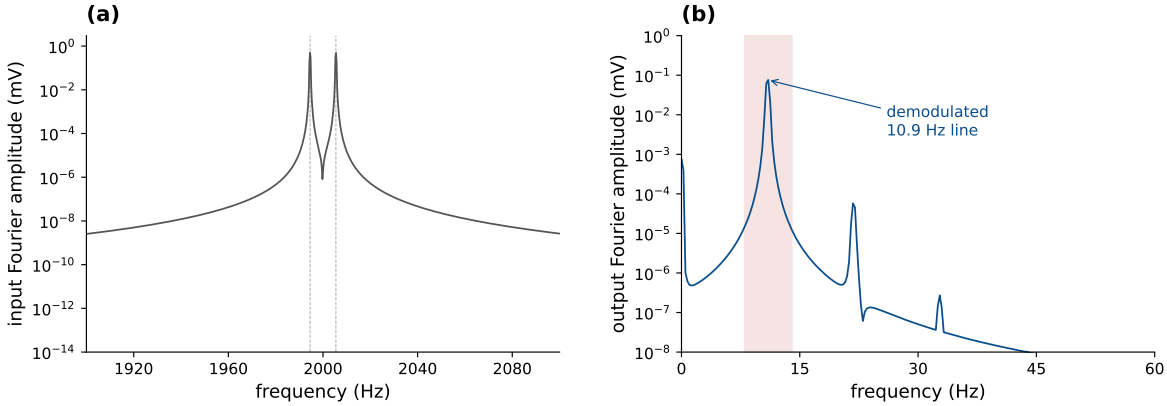


Figure S6: Two-tone TI at a kHz carrier (Jansen–Rit, assumed fast access). Two equal tones at $f_1 = 1994.543$ and $f_2 = 2005.457$ Hz, beating at 10.914 Hz, pass through the assumed low-pass ($\tau = 0.2$ ms) into the recurrent column. *(a)* The input amplitude spectrum has lines only at f_1 and f_2; in the 8–14 Hz band its maximum is 2.2×10^{-13} mV. *(b)* The output carries a demodulated line at the beat, 10.9 Hz.

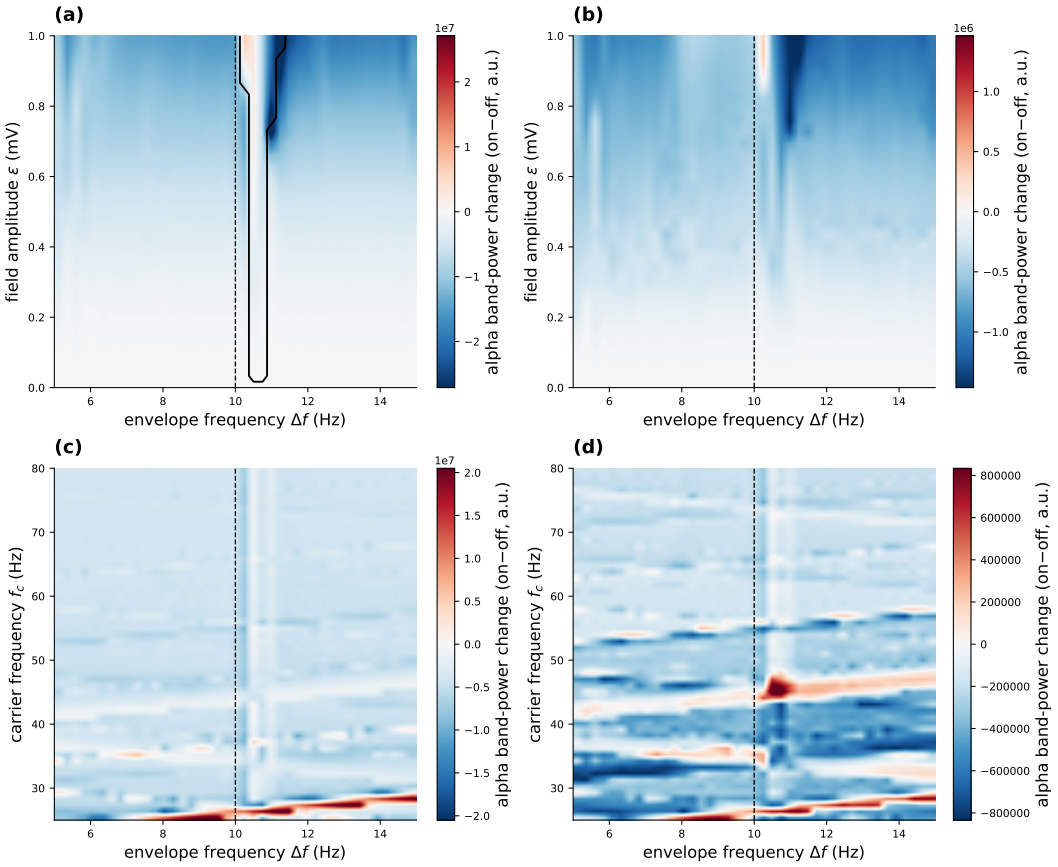


Figure S7: LaNMM frequency selection: two-tone TI field delivered directly to the deep P_1 (alpha) loop. As Fig. S1 but with the field on P_1. The deep loop shows frequency dominance at smaller fields, so the amplitude range is $\varepsilon \leq 1$ mV here against 3 mV there. These finite-amplitude maps extend beyond the perturbative regime of the theory. Color: signed change of alpha-band power from the $\varepsilon = 0$ baseline, zero-centered; most of the grid is suppression. *(a,b)* P_1 and P_2 enhancement vs. ε and Δf; the black contour in *(a)* outlines the frequency-dominance set $|f_{\text{out}} - \Delta f| < 0.3$ Hz (Appendix F), computed from the dominant output frequency because band power alone cannot separate a driven from an autonomous rhythm; the $\varepsilon = 0$ row is excluded (resolution floor 0.5 Hz). The set widens to 1.25 Hz at $\varepsilon = 1$ mV. *(c,d)* Enhancement vs. carrier frequency and Δf at $\varepsilon = 0.5$ mV. The ridge near alpha persists across the carrier band and includes the autonomous rhythm. A separate lock-in comparison away from alpha, including its window and background controls, is given in Table 3; these panels show band-power changes.

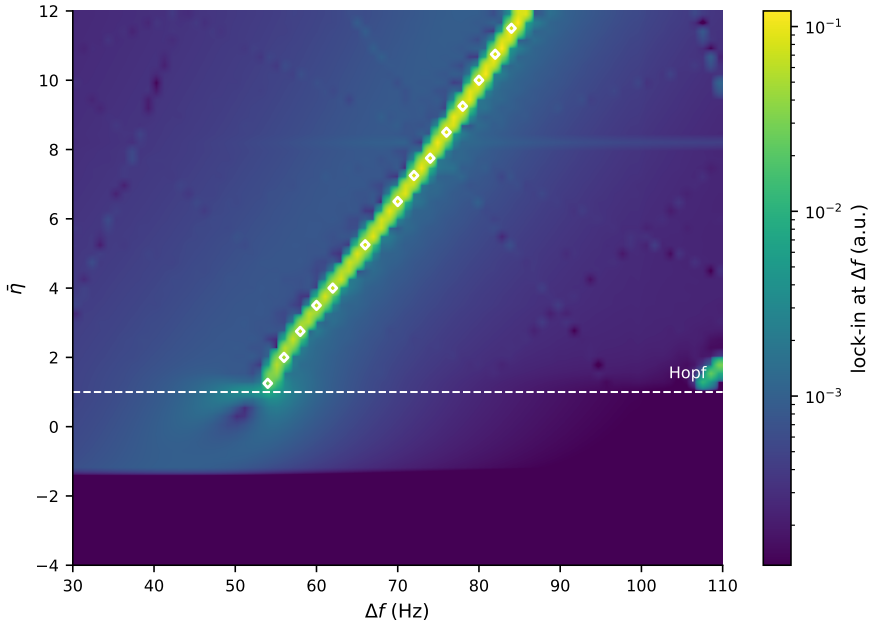


Figure S8: NMM2 PING resonance map (exact mean field). Demodulated response over the (Δf, background drive $\bar{\eta}$) plane, log color scale. The ridge follows the gamma natural frequency, with f_0 rising with $\bar{\eta}$ for these parameters. White markers outline the frequency-dominance set (Appendix F), drawn only above the Hopf (dashed line) where there is an autonomous rhythm to capture; below it the same color field is a forced response. The markers do not establish phase capture. Lock-in amplitude cannot separate driven and autonomous contributions, since both can place power at Δf.

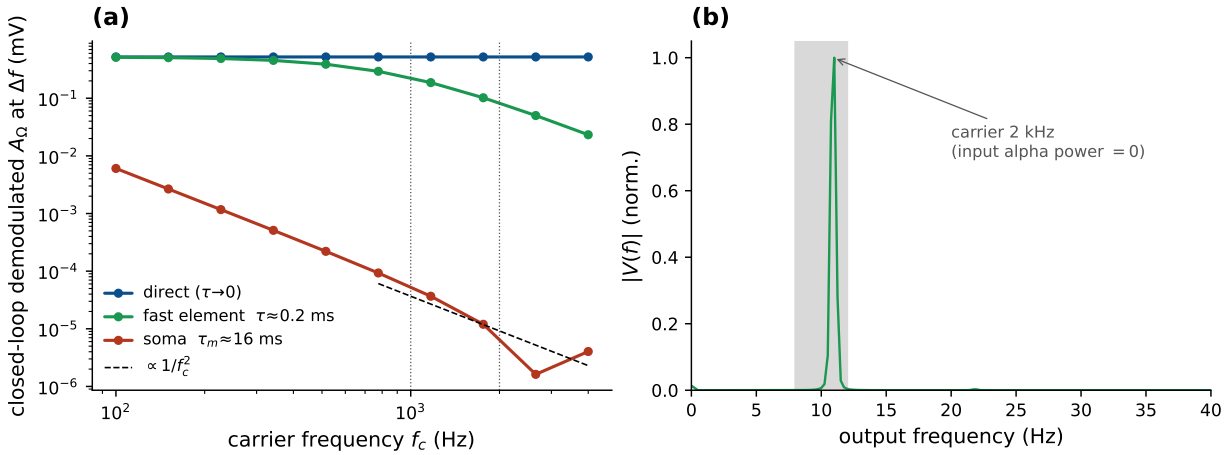


Figure S9: Demodulation at a kHz carrier (Jansen–Rit, closed loop). The applied field passes through an explicit membrane filter before the sigmoid. *(a)* Closed-loop demodulated A_Ω at $\Delta f = f_0$ vs. carrier f_c for three access models: direct ($\tau \to 0$, carrier-independent), fast ($\tau = 0.2$ ms), and somatic ($\tau_m = 16$ ms; $1/f_c^2$ roll-off). The fast time is assumed rather than assigned to a measured compartment. *(b)* Output spectrum for a 2 kHz carrier delivered through the fast element: an alpha line at f_0, synthesized by the column although the input carries no alpha power.

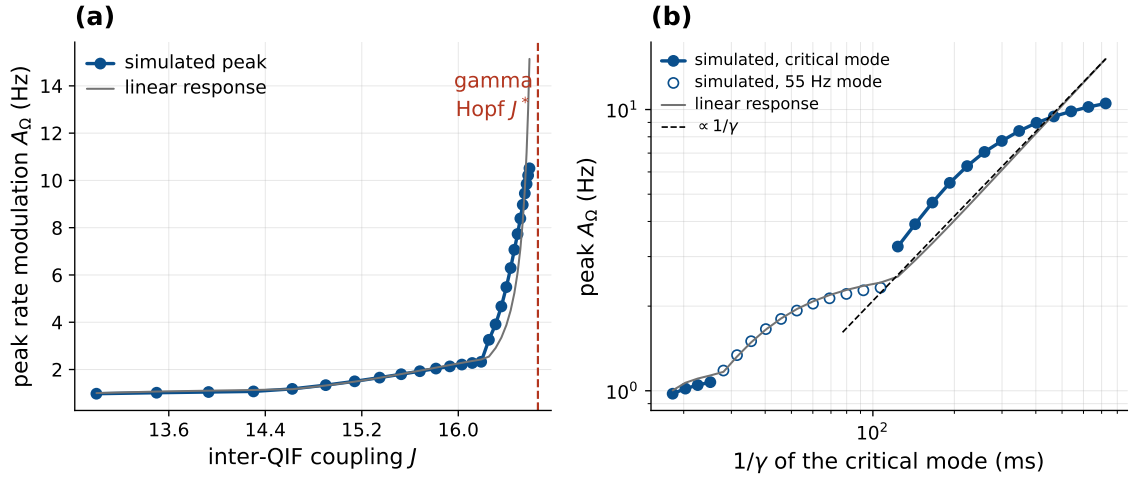


Figure S10: Coupling controls the amplification in the exact mean field (NMM2 PING, gamma). J is the inter-QIF synaptic weight; the sweep is unpinned at $\bar{\eta} = 0$ and uses the AM surrogate with $A_f = 8$. *(a)* Simulated peak of r_E (blue) and the linear reference (gray), scaled to the median measured gain over $14.5 < J < 16$. The reference includes the second-order carrier source; its nearby-carrier approximation changes the AM peak gain by less than 0.1%. The red dashed line marks the field-free Hopf. *(b)* The same responses against inverse damping of the critical mode, tracked continuously from the Hopf. Symbols are filled where that mode carries the linear peak and open where the second mode near 55 Hz does ($14.6 \lesssim J \lesssim 16.2$). The dashed $1/\gamma$ guide is anchored at the last linear-response point. The critical-mode gain follows it near the Hopf; the simulated response first exceeds and then falls below it at this amplitude. At a 50.5 Hz beat, the endpoint is attracting for both waveforms: the AM multiplier is 0.034 over 2 s, and two-tone perturbations decay at -0.0014 ms^{-1}. Both coupling and rectifier are derived from the QIF network.

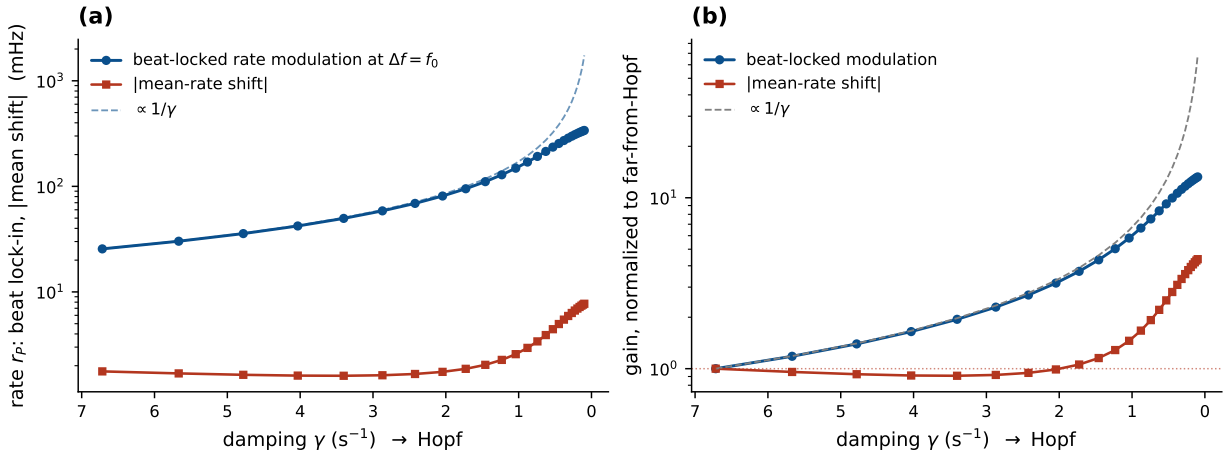


Figure S11: Envelope-locked modulation against mean-rate shift (Jansen–Rit, detection pinned). As the column is driven toward the Hopf (from the stable side, $\gamma > 0$) through the coupling at fixed $\sigma''(v^*)$, both quantities are read from the pyramidal rate $r_P = \sigma(v + s)$, in mHz. *(a)* The beat-locked modulation at $\Delta f = f_0$ (blue) is resonantly amplified and tracks $1/\gamma$ (dashed) before saturating; the magnitude of the mean-rate shift (red) stays at a few mHz and rises near the Hopf. *(b)* Both normalized to their far-from-Hopf values (log scale): the modulation grows 13.2× and the mean-shift magnitude 4.4× across the sweep. The input has two genuine tones. Modulation exceeds the mean shift by 14.5× far from the Hopf; the ratio peaks at 58× near $\gamma = 1$ s^{-1} and falls to 44× at the nearest point as the amplified oscillation rectifies. The AM surrogate gives 19, 66, and 45×, respectively.

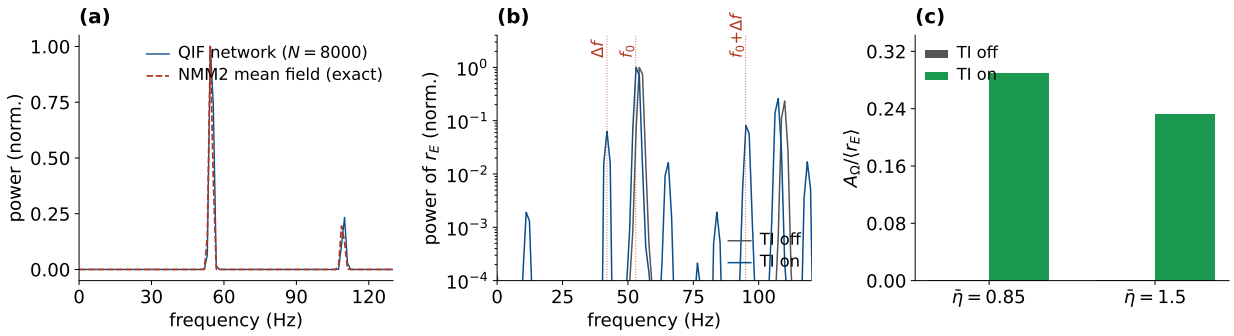


Figure S12: Beat-locked modulation and mean-rate change in the QIF network. *(a)* Field-free power spectrum of $r_E(t)$, normalized: the spiking network ($N = 8000$) matches the exact NMM2 mean field in gamma frequency (∼55 Hz); a normalized spectrum does not test the mean rate. *(b)* Intermodulation above the Hopf (TI off vs. on, log power): the nonlinearity mixes the imposed $\Delta f = 42$ Hz beat with the persisting gamma near 53–54 Hz, raising the direct Δf line and the sum sideband near 95 Hz, both absent with TI off. These sidebands establish nonlinear mixing; they do not uniquely identify gamma-amplitude gating. *(c)* The Δf modulation depth $A_\Omega/\langle r_E \rangle$, off vs. on: below 0.003 rising to 0.29 ($\bar{\eta} = 0.85$) and 0.23 ($\bar{\eta} = 1.5$), while the mean rate rises by about 8–11%. The labels identify field-free states; both field-on responses contain gamma activity beyond the beat component. Spectra use the same post-settling windows as the rate measurements and are normalized to each spectrum's maximum. Modulation depth is reported rather than the off→on lock-in ratio, whose denominator is the field-free floor.

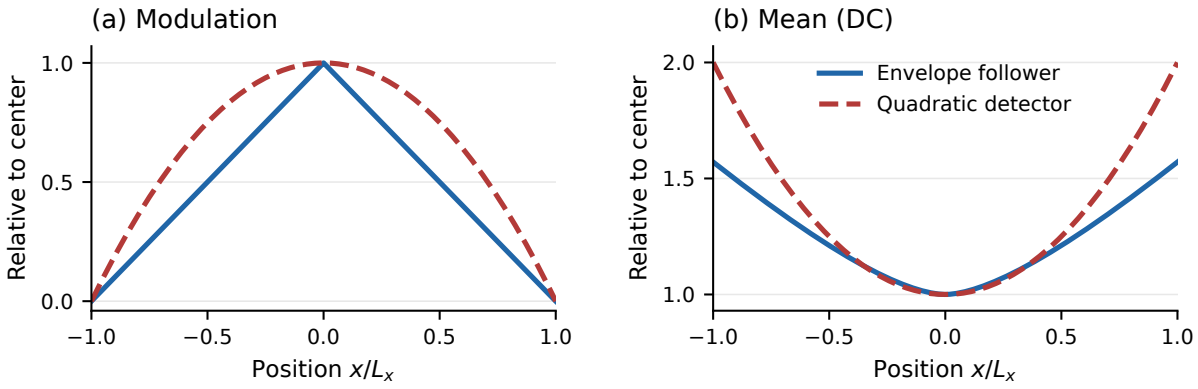


Figure S13: Envelope and quadratic spatial maps. Illustrative line profile: $\varepsilon_1(x) = \varepsilon_0(1 + x/L_x)$ and $\varepsilon_2(x) = \varepsilon_0(1 - x/L_x)$ on $-L_x \le x \le L_x$, with polarization scale $\varepsilon_0 > 0$ (mV) and length $L_x > 0$ (m). Each curve is divided by its own value at $x = 0$. *(a)* Modulation is $1 - |x/L_x|$ for envelope following and $1 - (x/L_x)^2$ for quadratic detection: both peak centrally, but the quadratic profile is broader here. *(b)* Both means increase toward the edges. This is a line profile, not a head model or a demonstration of three-dimensional focusing.

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